

SOC ANALYST

LEARNING NOTEBOOK

From Beginner to Job Ready



12 MODULES
15-DAY LEARNING PLAN



FOCUS



DETECT



INVESTIGATE



RESPOND



IMPROVE

Prepared by blueteamers.io



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Day 1
Module 1

SOC ANALYST ROLE & SOC ENVIRONMENT



1. What is SOC?

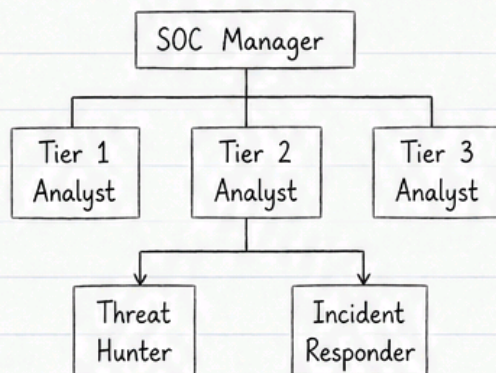
- SOC (Security Operations Center) is the heart of an organization's cyber defense.
- A centralized team that continuously monitors, detects, analyzes, and responds to security threats.

2. SOC Objectives

- Monitor organization's systems, networks & logs 24/7
- Detect and analyze security incidents
- Respond quickly to minimize impact
- Prevent future attacks through continuous improvement
- Protect confidentiality, integrity & availability



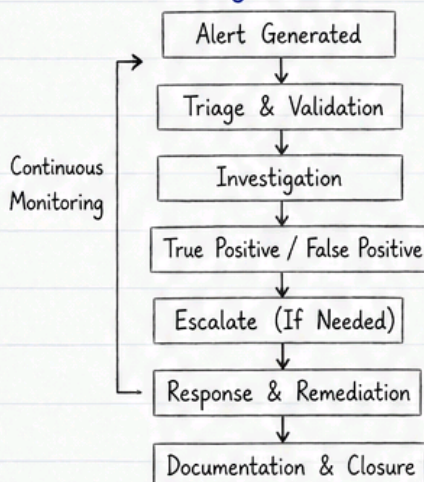
3. SOC Team Structure



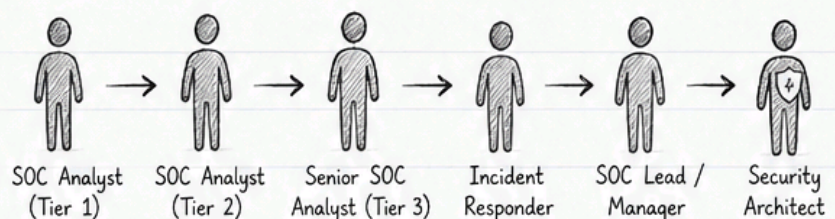
4. Daily Responsibilities (Tier 1 SOC Analyst)

- Monitor alerts from SIEM, EDR, IDS, Firewall, etc.
- Triage and validate alerts (True Positive / False Positive)
- Investigate incidents and collect evidence
- Escalate critical incidents to Tier 2
- Document all findings in ticketing system
- Communicate with teams when required
- Follow escalation and response procedures

5. SOC Working Model



6. Career Path



7. Key Skills for SOC Analyst

- | | |
|-------------------------------------|-----------------------|
| • Log Analysis | • Incident Response |
| • Network Fundamentals | • Communication |
| • Windows / Linux Basics | • Problem Solving |
| • SIEM Tools (Splunk, QRadar, etc.) | • Attention to Detail |
| • Threat Intelligence | • Continuous Learning |

NETWORK TRAFFIC ANALYSIS

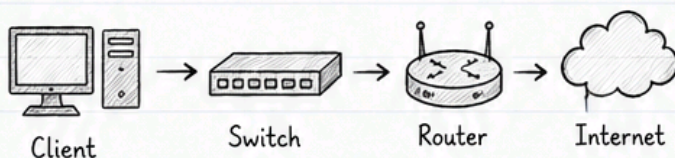
1. Why Network Traffic Analysis?

- Understand how data flows in a network
- Identify suspicious activity and attacks
- Troubleshoot network issues
- Essential skill for SOC Analysts

3. TCP/IP Model (4 Layers)

4	Application Layer	HTTP, DNS, SSH, FTP, SMTP
3	Transport Layer	TCP, UDP
2	Internet Layer	IP, ICMP
1	Network Access Layer	Ethernet, Wi-Fi

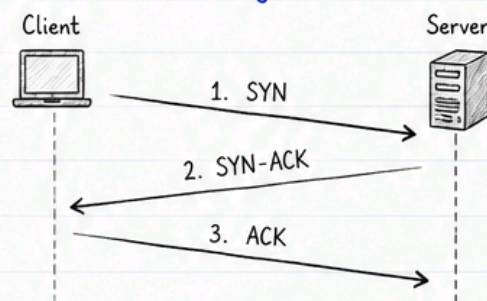
4. Basic Network Communication Flow



2. OSI Model (7 Layers)

7	Application Layer	HTTP, DNS, FTP, SMTP
6	Presentation Layer	SSL/TLS, Encoding, Encryption
5	Session Layer	Session Establishment, Mgmt
4	Transport Layer	TCP, UDP
3	Network Layer	IP, ICMP, Routing
2	Data Link Layer	MAC Address, Ethernet
1	Physical Layer	Bits, Cables, Signals

5. TCP Three-Way Handshake



6. Important Protocols

- DNS - Domain Name System (Port 53)
- HTTP - HyperText Transfer Protocol (Port 80)
- HTTPS - Secure HTTP (Port 443)
- SSH - Secure Shell (Port 22)
- FTP - File Transfer Protocol (Port 21)
- SMTP - Simple Mail Transfer Protocol (Port 25)
- ICMP - Internet Control Message Protocol

8. Key Network Concepts

- **IP Address** - Logical address (IPv4 / IPv6)
Example: 192.168.1.10
- **MAC Address** - Physical address of device
Example: 00:1A:2B:3C:4D:5E
- **Packet** - Unit of data in transit
- **Latency** - Time taken for data to travel
- **Bandwidth** - Capacity of the network

7. Common Ports Cheat Sheet

Port	Protocol	Service	Purpose
20 / 21	TCP	FTP	File Transfer
22	TCP	SSH	Secure Remote Access
23	TCP	Telnet	Remote Access (Insecure)
25	TCP	SMTP	Email Sending
53	UDP/TCP	DNS	Domain Name Resolution
80	TCP	HTTP	Web Traffic
110	TCP	POP3	Email Retrieval
143	TCP	IMAP	Email Retrieval
443	TCP	HTTPS	Secure Web Traffic
3389	TCP	RDP	Remote Desktop

9. Tools for Traffic Analysis


Wireshark
(Packet Capture & Analysis)


tcpdump
(Command Line Packet Capture)


nslookup / dig
(DNS Query Tools)

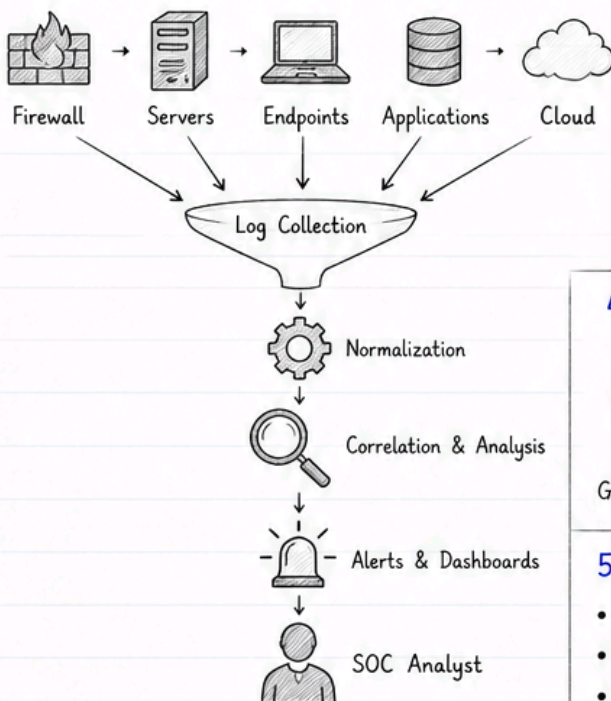

Netstat
(Network Connections & Statistics)


iftop
(Real-time Bandwidth Monitoring)

1. What is SIEM?

- SIEM (Security Information and Event Management) helps organizations detect, investigate and respond to security threats.
- It collects, normalizes, correlates and analyzes logs from multiple sources in real-time.
- Acts as the central hub for security visibility and monitoring.

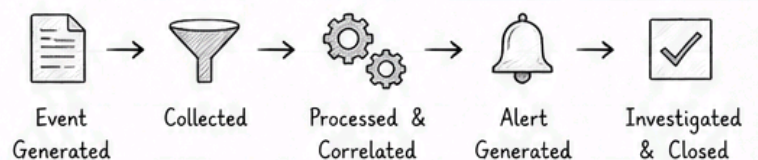
2. How SIEM Works? (Log Flow)



3. Key SIEM Components

	Log Collection	Collect logs from various sources (devices, apps, systems, cloud).
	Normalization	Convert logs into a common format for better analysis.
	Correlation	Identify patterns and link events to detect threats.
	Alerting	Generate alerts when suspicious activities are detected.
	Dashboards	Visualize security events and metrics.
	Reporting	Create reports for compliance, incidents and audits.

4. Event Lifecycle in SIEM



5. Common Use Cases

- Detect brute force login attempts
- Identify malware infections
- Detect data exfiltration
- Monitor privileged user activity
- Detect policy violations
- Compliance monitoring & reporting



6. Popular SIEM Solutions

SIEM	Key Features
Splunk	Powerful search, dashboards, alerts, integrations
IBM QRadar	Advanced correlation, offense management, threat intel
Microsoft Sentinel	Cloud-native, UEBA, SOAR integration, Azure native
Wazuh	Open-source, HIDS, log analysis, vulnerability detection
LogRhythm	UEBA, threat detection, compliance reporting

7. Basic Search Example (Splunk)

```

index=security sourcetype=auth
| where status="failed"
| stats count by user, src_ip
| sort -count
| head 10
  
```



Good queries + Better data =
Better detections

8. Best Practices

- ☒ Collect relevant logs from all critical sources
- ☒ Keep time synchronized (NTP)
- ☒ Create meaningful correlation rules
- ☒ Tune rules to reduce false positives
- ☒ Continuously review and improve dashboards

ENDPOINT INVESTIGATION

1. What is Endpoint?

- Endpoints are devices that connect to the network.
- Examples: Laptops, Desktops, Servers, Mobile Devices.
- Attackers often target endpoints to gain initial access to the network.



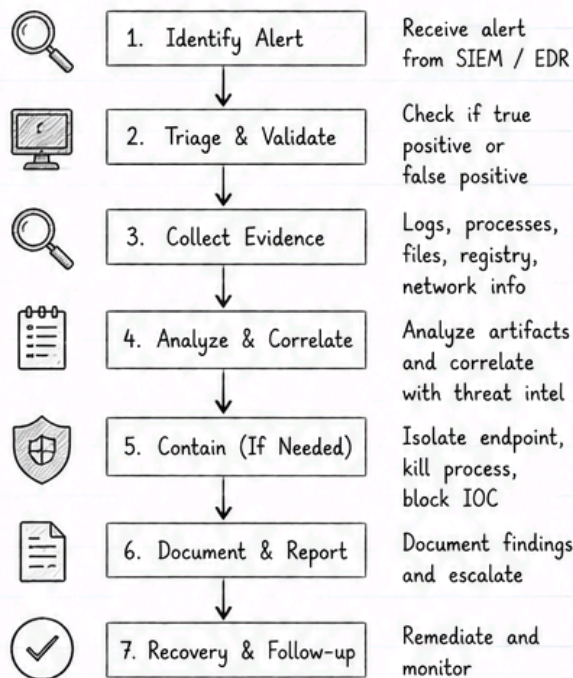
2. Endpoint Security Tools

- EDR (Endpoint Detection & Response)
- XDR (Extended Detection & Response)
- Antivirus / Anti-malware
- Host Firewall
- Device Control
- Application Control

3. Key Areas to Investigate on an Endpoint

<u>Processes</u>	<u>Services</u>	<u>Registry</u>	<u>Startup Items</u>	<u>Event Logs</u>
- Running processes - Suspicious processes - Process tree - High CPU/Memory usage	- Running services - Unusual services - Service startup type - Service binaries	- Run/RunOnce keys - Startup folders - Policies - Recently modified keys	- Startup programs - Scheduled tasks - Logon scripts - Browser extensions	- Windows Event Logs - Security logs - System logs - Application logs

4. Basic Investigation Workflow



5. Important Artifacts

Artifact	Purpose
Process List	Identify malicious or unusual processes
Network Connections	Check suspicious outbound connections
Registry Keys	Find persistence mechanisms
Event Logs	Review user activity and system events
Scheduled Tasks	Detect malicious scheduled tasks
Startup Items	Identify auto-start persistence
File Hashes	Verify file reputation
Browser History	Track user activity / malicious URLs

6. Useful Tools for Endpoint Investigation



7. Investigation Checklist (L1 SOC Analyst)

- | | |
|--|---|
| ☐ Review alert details | ☐ Review important registry keys |
| ☐ Identify affected endpoint | ☐ Check event logs (Security, System, Application) |
| ☐ Check running processes | ☐ Validate artifacts with threat intel |
| ☐ Review network connections | ☐ Document findings |
| ☐ Check startup items & scheduled tasks | ☐ Escalate if necessary |



PHISHING & EMAIL ANALYSIS

1. What is Phishing?

- Phishing is a social engineering attack where attackers trick users into revealing sensitive information.
- Attackers often use fake emails, malicious links or attachments.
- Goal: Steal credentials, install malware or gain unauthorized access.



2. Email Analysis Checklist

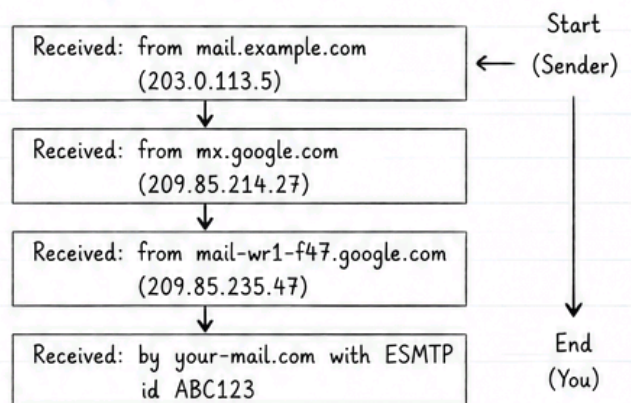
- ☐ Check email header
- ☐ Verify sender address carefully
- ☐ Look for suspicious links
- ☐ Analyze attachments
- ☐ Check SPF, DKIM, DMARC
- ☐ Check sender reputation
- ☐ Sandbox unknown files/URLs
- ☐ Report and escalate



3. Email Header - Key Fields

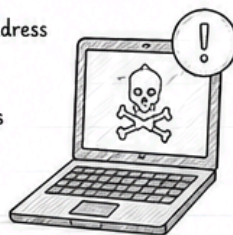
From	Shows who the email claims to be from.
Return-Path	Shows the actual email origin (used for SPF).
Received	Shows the path the email took to reach you.
SPF	Checks if the sending server is authorized.
DKIM	Verifies email integrity using digital signature.
DMARC	Policy for handling failed SPF/DKIM emails.
Message-ID	Unique ID of the email.

Example Email Header Flow



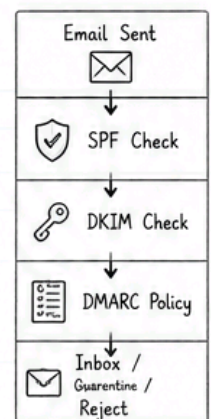
4. Common Phishing Indicators

- Generic greetings (Dear User, Dear Customer)
- Urgent or threatening language
- Mismatched or suspicious sender address
- Hover links point to different URL
- Poor grammar and spelling mistakes
- Unexpected attachments or links
- Requests for sensitive information



5. SPF, DKIM, DMARC - Explained

- SPF** (Sender Policy Framework): Verifies if the sending IP is authorized.
- DKIM** (DomainKeys Identified Mail): Verifies email integrity using digital signature.
- DMARC** (Domain-based Message Authentication, Reporting & Conformance): Policy to handle failed SPF/DKIM emails (None / Quarantine / Reject).



6. Attachments & Links - Safe Analysis

Attachments

- Check file extension
- Scan with AV
- Analyze in sandbox
- Never open in production environment



Links

- Hover to preview URL
- Check domain reputation
- Use URL scanning tools (VirusTotal, URLVoid)
- Open in sandbox / isolated browser



7. Useful Tools



VirusTotal
(File/URL Scanning)



MxToolbox
(Header Check)



Mail-Tester
(Spam Score)



Any.Run
(Sandbox Analysis)

8. Best Practices



Verify before you trust.



Analyze everything.



If in doubt, treat as threat.



Report suspicious emails.



Stay aware, stay secure.

INCIDENT HANDLING & REPORTING

1. What is Incident Handling?

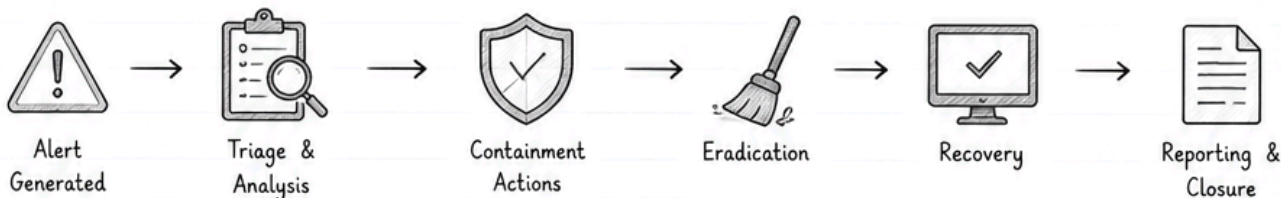
- Incident Handling is the process of identifying, managing and resolving security incidents.
- Goal is to minimize damage, recover quickly and prevent future incidents.
- A well-defined process ensures consistency and reduces response time.



2. Incident Lifecycle (NIST)

① Preparation	Prepare people, process, tools and playbooks.	
② Identification	Detect and analyze potential incidents.	
③ Containment	Limit the impact and stop the spread.	
④ Eradication	Remove the root cause and threats.	
⑤ Recovery	Restore systems and monitor for stability.	
⑥ Lessons Learned	Review the incident and improve process.	

3. Incident Handling Flow (Overview)



4. Severity Levels

Level	Severity	Description	Example
1	Critical	High impact, business disruption, security breach	Ransomware, Data Exfiltration
2	High	Potential for high impact if not resolved	Privilege escalation, Malware detected
3	Medium	Moderate impact, limited risk	Policy violation, Suspicious activity
4	Low	Low impact, informational alerts	Failed login, Port scan

5. Ticket / Report Information

- Incident ID
- Date & Time
- Alert Source
- Affected Systems / Users
- Incident Summary
- Impact (What happened)
- Actions Taken
- Current Status
- Escalated To
- Evidence / Artifacts
- Final Resolution
- Lessons Learned



6. Escalation Matrix (Example)

Severity	L1 Analyst	L2 Analyst	Manager	IR Team
Critical	✓	✓	✓	✓
High	✓	✓	✓	✓
Medium	✓	✓ (If needed)	✓ (If needed)	-
Low	✓	-	-	-

Escalate quickly, communicate clearly, document everything.

7. Incident Response Golden Rules

- Act fast, but don't panic.
- Verify the alert before taking action.
- Contain first, investigate next.
- Preserve evidence before making changes.
- Communicate every step.
- Document everything.
- Learn and improve after every incident.



8. Common Mistakes to Avoid

Ignoring alerts or delaying response	Not collecting enough evidence	Deleting logs or evidence accidentally	Poor communication	Incomplete documentation
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CLOUD SECURITY MONITORING

1. What is Cloud Security?

- Cloud Security refers to the policies, technologies, controls and practices used to protect data, applications and infrastructure in the cloud.
- The goal is to ensure confidentiality, integrity and availability of resources.
- Cloud environments are dynamic, scalable and shared – security must be continuous.

2. Shared Responsibility Model

Provider (e.g., AWS, Azure, GCP)	Customer (You)
• Physical Infrastructure • Network • Compute • Storage • Platform Services	• Data • Identity & Access Management • OS, Applications • Network Security (Security Groups) • Configuration • Monitoring & Logging

Security "IN" the cloud is your responsibility.

3. Major Cloud Platforms



- CloudTrail
- CloudWatch
- IAM
- VPC, Security Groups



- Azure Activity Logs
- Azure Monitor
- Azure AD (IAM)
- NSG (Network Security Group)



- Cloud Audit Logs
- Cloud Monitoring
- IAM
- VPC, Firewall Rules

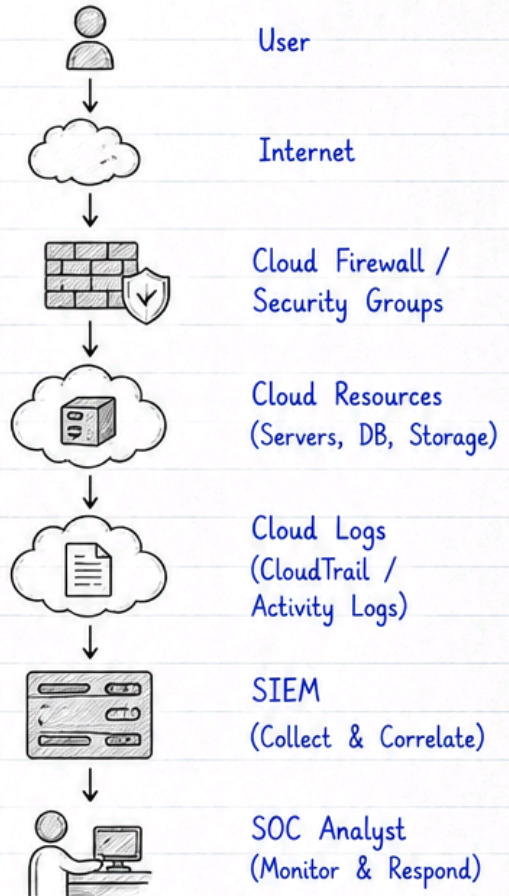
4. Key Cloud Security Concepts

- IAM – Manage users, roles and permissions.
- Least Privilege – Give minimum access required.
- Encryption – Encrypt data at rest and in transit.
- Logging – Enable logging for all services.
- Segmentation – Use VPCs and Security Groups.
- Monitoring – Continuously monitor activities.
- Backup – Regular backups and versioning.



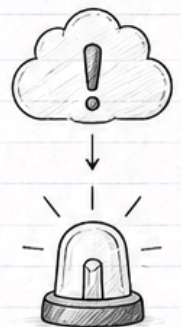
Cloud ≠ Automatically Secure
Misconfiguration is the #1 cause of cloud breaches.

5. Cloud Monitoring Architecture



6. Common Cloud Alerts

- Root / Admin login from new location
- Unusual API calls or service usage
- Creation of new IAM users / roles
- Security group rule modification
- Data exfiltration (large outbound transfer)
- Disabled logging or monitoring
- Privilege escalation attempts
- Unusual instance / resource creation



7. Best Practices

- ✓ Enable logging for all cloud services.
- ✓ Use IAM roles – avoid using root account.
- ✓ Apply least privilege access.
- ✓ Monitor and alert on suspicious activities.
- ✓ Encrypt data at rest and in transit.
- ✓ Regularly review permissions and configurations.
- ✓ Use Cloud Security Posture Management (CSPM).
- ✓ Backup critical data regularly.



1. What is Threat Intelligence?

- Threat Intelligence is evidence-based knowledge about existing or emerging threats.
- Helps organizations understand attackers, their motives, tools and TTPs.
- Enables proactive defense and better decision-making.



Think Like an Attacker,
Defend Like a Pro!

2. IOC vs IOA

IOC (Indicator of Compromise)	IOA (Indicator of Attack)
- Evidence that a breach has already occurred. - Specific and known. - Example: Malicious IP, Hash, Domain, File.	- Behavior that indicates an attack may be happening. - Patterns and anomalies. - Example: Abnormal login, Suspicious process.



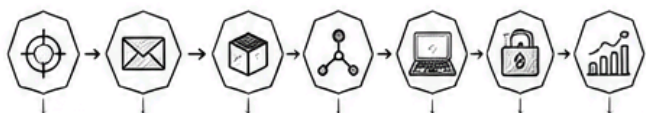
3. MITRE ATT&CK Framework

- A knowledge base of adversary Tactics, Techniques and Procedures (TTPs).
- Used worldwide to model attacks and improve defenses.
- 14 Tactics in the ATT&CK Matrix.

Initial Access	Execution	Persistence	Privilege Escalation
Defense Evasion	Credential Access	Discovery	Lateral Movement
Collection	Command & Control	Exfiltration	Impact

Use MITRE to map alerts and understand attacker behavior.

4. Cyber Kill Chain



1. Recon 2. Weaponize 3. Deliver 4. Exploit 5. Install 6. C2 7. Actions on Objectives



Understand each stage to detect and break the attack early.

5. Threat Intelligence Process



Threat Feeds

Collect data from different sources (Open/Commercial).



Collect & Process

Normalize, deduplicate and enrich the data.



Analyze & Validate

Identify relevance, verify and extract IOCs/IOAs.



Distribute

Share intelligence with teams and tools.



Detect & Respond

Use intel in SIEM/EDR/Firewall to detect and respond.

Threat Intelligence Sources

- Open Source (OSINT)
- Commercial Feeds
- ISAC/ISAO
- Dark Web
- Malware Analysis
- Internal Logs
- Honeypots
- Sandboxes

6. Threat Hunting Process



- Hypothesis**
Create a theory about threat.
- Hunt**
Search across data sources.
- Investigate**
Validate findings and collect evidence.
- Report**
Document and share results.
- Improve**
Update rules, detection and knowledge.



Hunting is not a one-time task,
it's a continuous improvement cycle.

7. Useful Tools & Feeds



VirusTotal



AlienVault
OTX



AbuseIPDB



MISP



Recorded
Future



YARA



Sigma



Suricata



Zeek



Splunk

8. Best Practices

- Validate intelligence before use.
- Correlate with multiple sources.
- Keep threat feeds updated.
- Use automation for enrichment.
- Share intelligence across the organization.
- Hunt proactively, not just reactively.



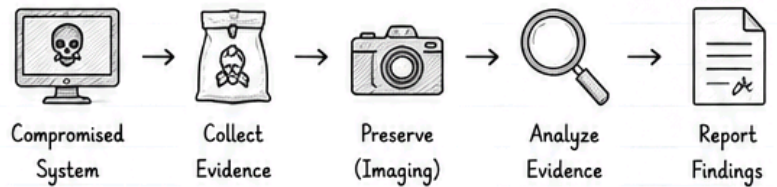
1. What is Digital Forensics?

- Digital Forensics is the process of identifying, preserving, analyzing and presenting digital evidence in a manner that is legally admissible.
- Goal: Find the truth from digital evidence while maintaining integrity and chain of custody.



Evidence speaks
Let the data tell the story.

2. Forensic Investigation Workflow

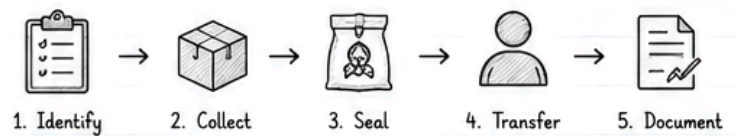


Always follow the process.
Never skip steps.

3. Important Evidence Sources

Evidence	What It Helps To Find
Memory (RAM)	Running processes, Malware, Encryption keys, Network connections
Hard Drive / SSD	Files, Deleted data, Documents, OS artifacts, Logs
Browser Artifacts	History, Cookies, Downloads, Saved credentials
Registry	System configurations, Persistence mechanisms
Event Logs	User activities, Logons, System errors, Timeline
Email & Chat Logs	Communication, Attachments, Sender/Receiver info
Removable Devices	Copied files, File transfer artifacts
Mobile Devices	Calls, Messages, Apps, Location, Photos

4. Chain of Custody



- Maintain a complete record of who collected the evidence, when, where, why and how it was handled.
- Break in the chain = Evidence can be rejected in court.
- Use evidence labels, signatures and timestamps.



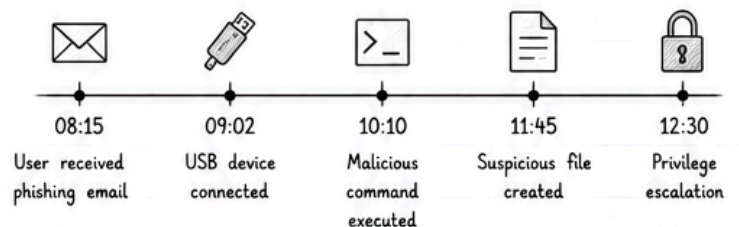
5. Common Forensic Tools

Autopsy (Disk Analysis)	FTK Imager (Imaging)	Volatility (Memory Analysis)	Wireshark (Network Analysis)
RegRipper (Registry Analysis)	KAPE (Artifact Collection)	Timeline Explorer (Timeline Analysis)	Bulk Extractor (Data Carving)

6. Key Artifacts to Look For

Artifact	Where to Find	Why It Matters
Recent Files	User Profile	Find recently accessed files
Prefetch Files	C:\Windows\Prefetch	Program execution history
Jump Lists	User Profile	Recently opened items
MRU Lists	Registry	Most Recently Used items
USB History	Registry	USB devices connected
User Accounts	System	Created accounts, usage
Scheduled Tasks	System	Persistence, Automation
ShimCache	Registry	Program execution history
Event Logs	Event Viewer	User activity timeline

7. Timeline Analysis (Build the Story)



Timeline helps you understand
what happened, when and how.

8. Forensic Best Practices



Always create forensic image, never analyze original data.



Use write blockers while imaging drives.



Document everything.



Maintain time synchronization (NTP).



Store evidence securely and limit access.



Follow legal guidelines and policies.



Verify findings and cross-check evidence.



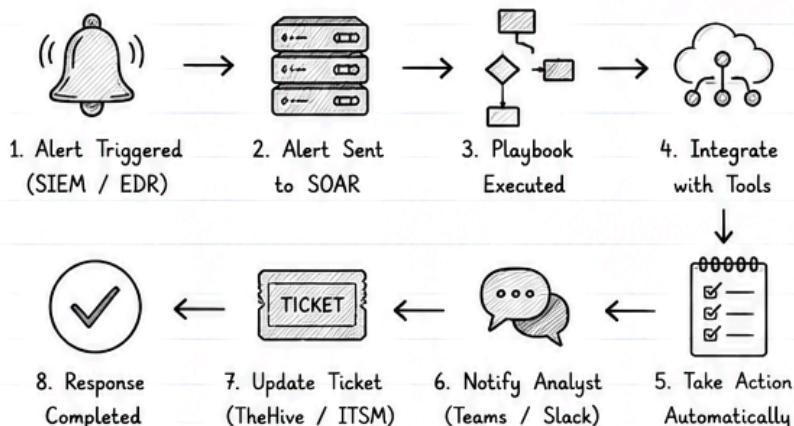
1. What is SOAR?

- SOAR (Security Orchestration, Automation and Response) combines people, processes and technologies to automate and orchestrate security operations.
- It helps SOC teams respond faster, reduce manual work and ensure consistency.
- Goal: Faster response, less noise, better accuracy, more time for hunting.



Automate the repeatable.
Orchestrate the complex.
Empower the analyst.

2. SOAR Automation Flow (High Level)



SOAR = Speed + Accuracy + Consistency

3. Key SOAR Components

Component	Description
Orchestration	Connects multiple tools and coordinates processes.
Automation	Executes repetitive tasks without human intervention.
Playbooks	Pre-defined workflows for common security scenarios.
Integration	Connects SIEM, EDR, TI, ITSM, Chat tools and more.
Reporting	Generate reports, metrics and audit trails.

4. Popular SOAR Platforms



TheHive

- Open Source
- Incident Response Platform
- Case Management
- Built-in MISP Integration



Shuffle

- Open Source
- Workflow Automation
- Drag & Drop Editor
- Strong Integrations
- Playbook Templates



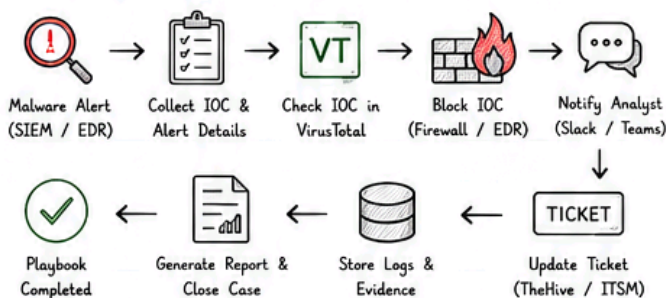
Tines

- Cloud SOAR
- Easy Automation
- Pre-built Actions
- Great for Small Teams



Choose the tool based on your use case, team size and environment.

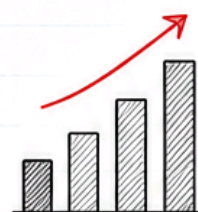
5. Example Playbook: Malware Detected



Automated response saves time and reduces human error.

6. Benefits of SOAR

- ✓ Reduce Mean Time to Respond (MTTR)
- ✓ Eliminate repetitive and manual tasks
- ✓ Improve accuracy and consistency
- ✓ Better integration between tools
- ✓ Free up analysts for hunting and investigation
- ✓ Scalable for growing SOC teams



SOAR does not replace analysts.
It makes analysts Super Analysts.

7. Common Use Cases

- Phishing email automation
- Malware analysis & response
- IOC enrichment & blocking
- User account lock / unlock
- Insider threat triage
- Vulnerability alert handling
- Data exfiltration alerts
- Threat intelligence enrichment
- Case management automation
- Compliance and reporting



8. Best Practices



- Start small - automate simple, high-impact tasks
- Test playbooks before using in production
- Use version control for playbooks
- Monitor and review automation regularly
- Keep playbooks updated with changing threats
- Document everything

1. What is Vulnerability Management?

- Vulnerability Management is the continuous process of identifying, evaluating, treating and reporting security vulnerabilities.
- Goal is to reduce the attack surface and minimize risk to the organization.
- It is an ongoing cycle, not a one-time task.



Find it. Fix it. Verify it.
Repeat it.

2. Vulnerability Management Lifecycle



1. Asset
Discovery



2. Vulnerability
Scanning



3. Risk
Assessment



4. Remediation
(Patch / Fix)



5. Verification
(Rescan)



6. Reporting
& Monitoring



A strong process today prevents a breach tomorrow.

3. Key Concepts

Concept	Description
Vulnerability	A weakness that can be exploited by an attacker.
Threat	A potential action or event that exploits a vulnerability.
Risk	Impact of a threat exploiting a vulnerability.
CVE	Common Vulnerabilities and Exposures - unique ID for vulnerabilities.
CVSS	Common Vulnerability Scoring System - standard to measure severity.
Remediation	Action taken to fix or mitigate a vulnerability.

4. CVSS Severity Levels

Severity	CVSS Score (Base)	Description	Action
Critical	9.0 - 10.0	Highly exploitable, severe impact	Fix Immediately
High	7.0 - 8.9	High impact, easy to exploit	Fix as soon as possible
Medium	4.0 - 6.9	Moderate impact, limited exploitability	Fix in normal cycle
Low	0.1 - 3.9	Low impact, hard to exploit	Fix when time permits
None	0.0	Informational no risk	No action required

5. Example CVE



- CVE ID : CVE-2023-12345
- Description : Remote Code Execution in XYZ Software
- CVSS Score : 9.8 (Critical)
- Affected : Version 2.1 and below
- Fix : Update to Version 2.2.1

6. Vulnerability Scanning Tools

nessus Nessus Popular scanner for networks and systems	Qualys Qualys Cloud based vulnerability management	RAPID7 Rapid7 InsightVM for continuous assessment	OpenVAS Open source vulnerability scanner	nexpose Nexpose Vulnerability management solution
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7. Remediation Workflow



1. Identify Vulnerability
2. Prioritize (Risk Based)
3. Apply Patch / Fix
4. Verify with Rescan
5. Update CMDB & Report



Patch early, patch smart, patch always.

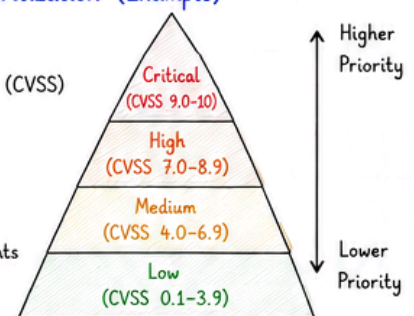
9. Best Practices

- ✓ Maintain accurate asset inventory
- ✓ Regularly scan internal & external assets
- ✓ Prioritize based on risk, not just CVSS score
- ✓ Apply patches in test environment first
- ✓ Verify remediation with re-scan
- ✓ Document and track all vulnerabilities
- ✓ Monitor for new vulnerabilities continuously



8. Risk Based Prioritization (Example)

- Asset Criticality
- Vulnerability Severity (CVSS)
- Exploit Availability
- Threat Intelligence
- Business Impact
- Regulatory Requirements



10. Common Vulnerability Types



Unpatched
Software



Injection
(SQL, Command)



XSS
Cross Site Scripting



Broken
Authentication



Misconfiguration



Sensitive Data
Exposure



Insecure Services
& Protocols



Insufficient Access
Control



1. MITRE ATT&CK OVERVIEW

- MITRE ATT&CK is a global knowledge base of adversary tactics and techniques.
- Helps SOC Analysts understand how attackers think and how to detect them early.



Understand the adversary.
Map the technique.
Stop the attack.

2. COMPLETE ATTACK FLOW



What Happens?

The attacker moves step by step inside the network, steals data and finally causes impact.



Key Point

Attackers follow a pattern.
We break the pattern.
Detect early,
Respond faster.
Prevent damage.

3. DETECTION OPPORTUNITIES



Windows Event Logs

Logon, Process Creation, File Access, Registry Changes.



Sysmon

Provides detailed visibility into process, network and file activity.



EDR

Detects suspicious behavior and blocks malicious activity.



SIEM Correlation

Correlate multi-source logs to detect complex attacks.



Network Logs

Detect C2 connections, data exfiltration and lateral movement.

4. COMMON ATT&CK TECHNIQUES

- Pass-the-Hash
- PowerShell Abuse
- Living off the Land (LOLBins)
- Scheduled Tasks
- Credential Dumping (Mimikatz)
- DLL Sideload
- WMI Abuse
- Fileless Malware



Remember

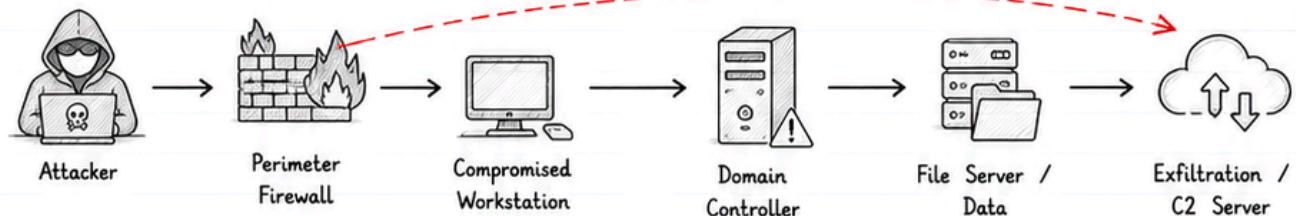
Attackers don't always use malware. They abuse legitimate tools and features.

Think Like an Analyst



If it's unusual, it's suspicious.

5. ATTACKER MOVEMENT EXAMPLE (PENCIL SKETCH)



Defend Early. Detect Fast.

Every second counts in cybersecurity. Your detection can prevent a disaster.

Pro Tips for SOC Analysts

- ✓ Know the attacker techniques.
- ✓ Monitor critical logs.
- ✓ Correlate, Validate, Escalate.
- ✓ Document everything.

Golden Rule

Detect early.
Contain quickly.
Learn always.





1. MITRE ATT&CK OVERVIEW

- MITRE ATT&CK is a global knowledge base of adversary tactics and techniques.
- It helps SOC Analysts understand how attackers think and how to detect them early.



Use ATT&CK to think like an attacker and defend like a pro.

3. DETECTION OPPORTUNITIES



Windows Event Logs

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Sysmon

Provides detailed visibility into process, network and file activity.



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Detects suspicious behavior and blocks malicious activity.



SIEM Correlation

Correlate multi-source logs to detect complex attacks.



Network Logs

Detect C2 connections, data exfiltration and lateral movement.

2. COMPLETE ATTACK FLOW (KILL CHAIN)



What Happens?

The attacker moves step by step inside the network, steals data, maintains access and finally causes impact.



Key Point

Attackers follow a pattern.

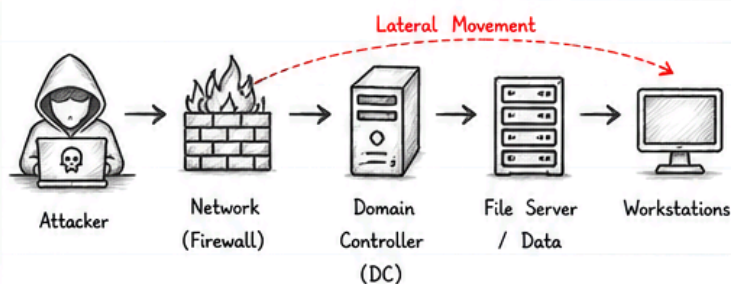
We break the pattern.

Detect early.
Respond faster.
Prevent damage.

4. COMMON ATT&CK TECHNIQUES (EXAMPLES)

- **Pass-the-Hash** - Uses stolen hashes to authenticate.
- **PowerShell Abuse** - Uses PowerShell for malicious activities.
- **Living off the Land (LOLBins)** - Uses legitimate tools for attacks.
- **Scheduled Tasks** - Maintains persistence using tasks.
- **Credential Dumping (Mimikatz)** - Extracts passwords from memory.
- **DLL Sideload** - Loads malicious DLL for execution.
- **WMI Abuse** - Uses WMI for remote execution and persistence.
- **Fileless Malware** - Runs in memory, leaves no files.

5. ATTACKER MOVEMENT EXAMPLE (PENCIL SKETCH)



Pro Tips for SOC Analysts

- ✓ Understand the adversary techniques.
- ✓ Map alerts with MITRE ATT&CK.
- ✓ Validate, Correlate and Investigate.
- ✓ Hunt proactively. Don't wait for alerts.
- ✓ Document everything.



Remember

Attackers only need one mistake.
Defenders must be ready every time.

How to Use MITRE ATT&CK?

- ★ Map alerts to ATT&CK techniques.
- ★ Improve detections using ATT&CK.
- ★ Understand attacker TTPs (Tactics, Techniques & Procedures).
- ★ Create better rules and use cases.
- ★ Train and build use-case scenarios.





Tools are important, but **understanding the workflow** is more important.



1. SIEM



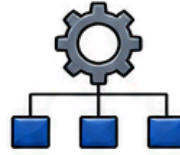
- **Splunk** – Log Management, Search, Dashboards
- **IBM QRadar** – Correlation, Offense Management
- **Microsoft Sentinel** – Cloud SIEM, Built for Azure
- **Wazuh** – Open Source SIEM, HIDS, Vulnerability Detection

2. EDR / XDR



- **Microsoft Defender** – Endpoint Protection, Threat Detection
- **CrowdStrike** – EDR, Threat Hunting, Incident Response
- **SentinelOne** – Autonomous Endpoint Protection
- **LimaCharlie** – Open Source EDR, Real-time Response

3. SOAR



- **Shuffle** – Automation & Orchestration
- **TheHive** – Incident Response Platform
- **Tines** – Cloud SOAR, Automation Made Simple

4. NETWORK ANALYSIS



- **Wireshark** – Packet Capture & Analysis
- **Zeek (Bro)** – Network Security Monitoring
- **Suricata** – IDS / IPS / NSM
- **tcpdump** – Packet Capture (CLI Tool)

5. THREAT INTELLIGENCE



- **VirusTotal** – File, URL, IP Analysis
- **AbuseIPDB** – IP Reputation Lookup
- **AlienVault OTX** – Threat Intel Feeds
- **MISP** – Threat Sharing Platform

6. EMAIL & WEB ANALYSIS



- **MXToolbox** – Email Header Analysis
- **Any.Run** – Interactive Malware Analysis
- **URLScan.io** – URL & Website Analysis
- **PhishTank** – Phishing Database

7. DFIR (DIGITAL FORENSICS)



- **Autopsy** – Forensic File Analysis
- **FTK Imager** – Disk Imaging Tool
- **Volatility** – Memory Forensics
- **KAPE** – Triage & Artifact Collection

8. VULNERABILITY MANAGEMENT



- **Nessus** – Vulnerability Scanner
- **OpenVAS** – Open Source Scanner
- **Qualys** – Cloud Based VM
- **Rapid7 InsightVM** – Risk Based Vulnerability Management

9. OTHER HELPFUL TOOLS

- **Sysinternals Suite** – Windows Troubleshooting
- **PowerShell** – Automation & Administration
- **Git & GitHub** – Version Control & Sharing
- **Notepad++ / VS Code** – Log Analysis & Scripting
- **Windows Event Viewer** – System & Application Logs
- **Mimikatz** – Credential Dumping (Authorized Use Only)

10. HOW TO CHOOSE THE RIGHT TOOL?

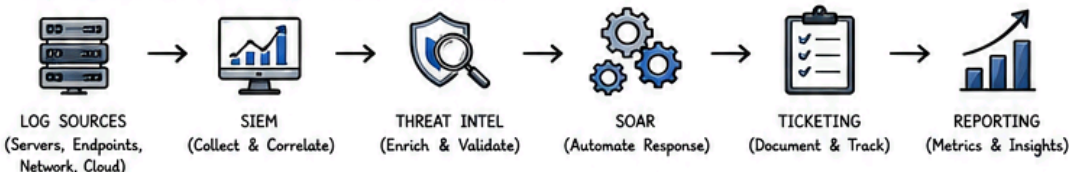
- ✓ Understand your requirement
- ✓ Know your environment (On-prem / Cloud)
- ✓ Start with free / open source tools
- ✓ Learn the workflow, not just the tool
- ✓ Practice with real scenarios
- ✓ Integrate tools for better visibility

REMEMBER

Tools help you find answers.

Skills help you analyze them.

TOOL ECOSYSTEM (HOW THEY WORK TOGETHER)



GOLDEN RULE

The best tool is the one you **understand and use effectively.**

PRO TIPS FOR SOC ANALYSTS

- ✓ Learn the purpose and workflow of each tool.
- ✓ Integrate tools to reduce alert fatigue.
- ✓ Automate repetitive tasks using SOAR.
- ✓ Keep your tools and knowledge up to date.
- ✓ Practice regularly in a lab environment.

WHAT MATTERS MOST?



- ✓ Detect early
- ✓ Investigate deeply
- ✓ Respond quickly
- ✓ Document everything
- ✓ Improve continuously

RECOMMENDED LAB PLATFORMS

- TryHackMe
- LetsDefend
- BlueTeamLabs Online
- Splunk Boss of the SOC
- Wazuh Lab
- DetectionLab



★ A structured workflow ensures **consistent**, **accurate** and **effective** incident handling.

WHAT IS THE SOC WORKFLOW?

A step-by-step process followed by SOC Analysts to investigate alerts, determine threats, respond quickly and document everything.



WORKFLOW PRINCIPLES



Be Thorough
Don't rush. Check every detail.



Be Objective
Follow evidence, not assumptions.

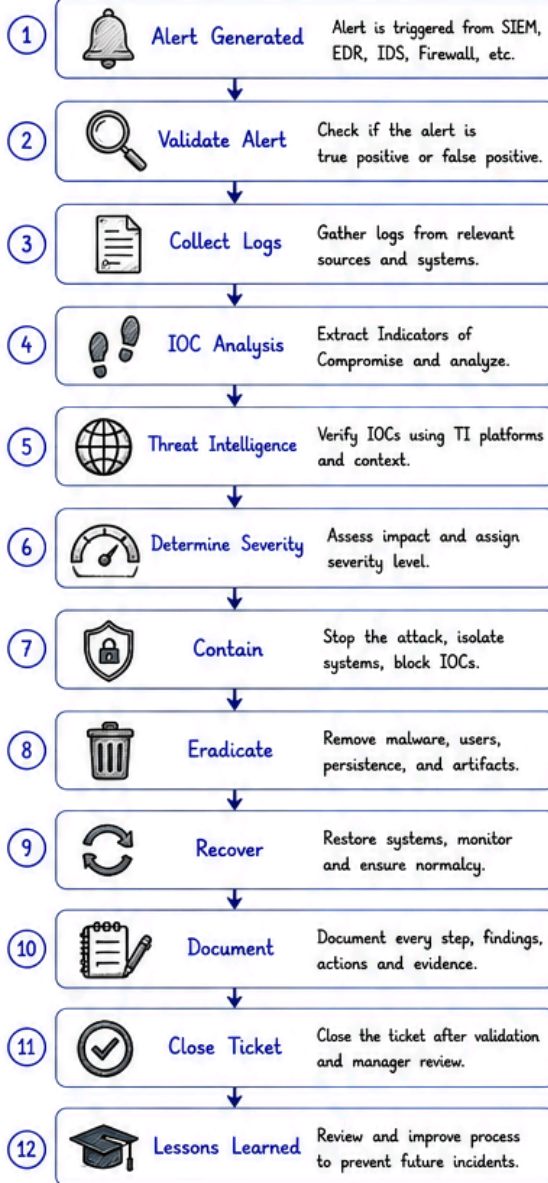


Be Consistent
Use the same process every time.



Be Communicative
Escalate and update clearly.

SOC INVESTIGATION WORKFLOW



L1 SOC ANALYST CHECKLIST



- ☐ Review the alert details
- ☐ Validate true or false positive
- ☐ Check affected host / user
- ☐ Collect logs from all sources
- ☐ Analyze logs and events
- ☐ Search IOCs in TI platforms
- ☐ Determine attack pattern
- ☐ Assign severity
- ☐ Escalate if required
- ☐ Take containment actions
- ☐ Document all findings
- ☐ Update ticket
- ☐ Communicate with team
- ☐ Close after validation



NEVER FORGET

Every alert tells a story. Your job is to understand the story, stop the attacker, and protect the organization.

KEY GOALS



- Detect Early
- Analyze Accurately
- Respond Quickly
- Minimize Impact
- Improve Continuously

COMMON MISTAKES



- ✗ Closing alerts too quickly
- ✗ Ignoring small indicators
- ✗ Not checking parent process
- ✗ Wrong severity assignment
- ✗ Poor documentation
- ✗ Not escalating critical alerts
- ✗ Assuming without evidence

BEST PRACTICES



- ✓ Follow the workflow strictly
- ✓ Use TI and contextual data
- ✓ Correlate across multiple sources
- ✓ Document every action
- ✓ Communicate and escalate
- ✓ Learn from every incident
- ✓ Automate repetitive tasks

TIP FOR SUCCESS



- Stay calm and follow the process.
- Use checklists and SOPs.
- Keep learning and practicing.
- Build home lab projects.
- Ask questions and collaborate.
- Curiosity + Discipline = Great Analyst



A great SOC Analyst doesn't just respond to alerts, They understand attackers, think critically and protect organizations.





Consistent Learning + Practical Skills + Right Mindset = **Successful SOC Analyst**



1. COMPLETE LEARNING ROADMAP



2. 30-DAY REVISION CHECKLIST

- ✓ Networking Concepts & Protocols
- ✓ Windows Logs & Event IDs
- ✓ Linux Commands & Logs
- ✓ SIEM Basics & Advanced Queries
- ✓ Log Sources & Normalization
- ✓ Alert Triage & Investigation
- ✓ Endpoint Monitoring & EDR
- ✓ Email Threats & Phishing Analysis
- ✓ Incident Handling Process
- ✓ Threat Intelligence
- ✓ Threat Hunting Techniques
- ✓ Cloud Security Monitoring
- ✓ SOAR Playbooks & Automation
- ✓ Digital Forensics Basics
- ✓ Vulnerability Scanning & Management
- ✓ MITRE ATT&CK Framework
- ✓ Reporting & Documentation
- ✓ Home Lab Projects
- ✓ Interview Preparation



SUCCESS FORMULA

Learn → Practice → Build →
Document → Share → Grow

3. INTERVIEW QUICK TIPS



Understand concepts, not just tools.



Explain your thought process clearly.



Show how you investigate step by step.



Practice writing clear incident reports.



Build a home lab and work on real projects.



Keep learning and stay updated.



Communication is a key skill for SOC Analysts.



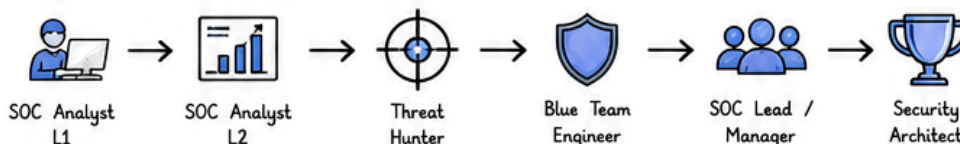
MINDSET FOR SUCCESS

- ✓ Be Curious
- ✓ Stay Consistent
- ✓ Never Stop Learning
- ✓ Help & Share Knowledge
- ✓ Think Like an Analyst

4. RECOMMENDED PRACTICE PLATFORMS

 TryHackMe Hands-on labs & learning paths.	 LetsDefend Blue Team labs & SOC challenges.	 BlueTeamLabs Online Realistic SOC scenarios.	 splunk Splunk Boss of the SOC (BOTS) Great for SIEM practice.	 DetectionLab Detection Engineering & Threat Hunting.	 Wazuh Lab SIEM + EDR Open Source Platform.	 AWS / Azure / TryCloud Cloud Security Practice Labs.
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5. CAREER PATH AFTER SOC ANALYST L1



REMEMBER

Your journey is unique. Focus on progress, not perfection.



A SOC Analyst doesn't just investigate alerts—they **understand** attacker behavior, **protect** organizations, and continuously **improve** their defenses.





This quiz is designed to help you revise key concepts from Networking, SIEM, EDR, Incident Response, Threat Hunting, Cloud Security, and more.



SECTION 1: MULTIPLE CHOICE QUESTIONS

- What is the primary goal of a SOC Analyst?
 - Develop software
 - Monitor and detect security threats
 - Manage hardware assets
 - Conduct penetration testing
- Which OSI layer does a firewall primarily operate on?
 - Layer 3
 - Layer 4
 - Layer 7
 - Layer 2
- Which SIEM tool is known for its correlation and offense management?
 - Splunk
 - Wireshark
 - QRadar
 - Nmap
- Which Windows Event ID indicates a successful logon?
 - 4624
 - 4625
 - 4688
 - 4719
- What is the purpose of EDR?
 - Monitor network traffic
 - Detect and respond to endpoint threats
 - Scan for vulnerabilities
 - Analyze email headers
- Which of the following is a common phishing indicator?
 - HTTPS in URL
 - Urgent action in email
 - Correct grammar
 - Known sender
- What is the first step in the incident response lifecycle?
 - Contain
 - Detect and Analyze
 - Recover
 - Document
- Which tool is used for network packet analysis?
 - Wireshark
 - Metasploit
 - John the Ripper
 - Burp Suite
- What is MITRE ATT&CK used for?
 - Vulnerability scanning
 - Mapping attacker techniques
 - Log management
 - Encryption
- Which cloud service model gives you the most control?
 - SaaS
 - PaaS
 - IaaS
 - FaaS



GOAL

Reinforce learning and build confidence for real-world SOC operations.



QUIZ INSTRUCTIONS

- ☒ Read each question carefully.
- ☒ Choose the best answer.
- ☒ Be honest with yourself.
- ☒ Review the explanations.
- ☒ Learn and improve!



SCORING GUIDE

14 - 16 : Excellent 🏆
11 - 13 : Good 👍
7 - 10 : Average 😊
0 - 6 : Needs Improvement 📉

Don't worry! Keep learning and keep practicing.



SECTION 2: TRUE OR FALSE

- A SIEM collects logs from multiple sources. _____
- EDR tools are used only for malware detection. _____
- Phishing emails always contain malicious attachments. _____
- MITRE ATT&CK helps in understanding attacker behavior. _____
- The OSI Model has 7 layers. _____
- Incident response ends after containment. _____
- Cloud security is the responsibility of the cloud provider. _____
- Threat hunting is a proactive security activity. _____



SECTION 3: MATCH THE FOLLOWING

Column A

- Event ID 4624
- Wireshark
- MITRE ATT&CK
- Phishing
- EDR
- DNS Tunneling
- TheHive
- Incident Response

Column B

- Packet Analysis Tool
- Successful Logon
- Attacker Technique Framework
- Deceptive Email
- Security Orchestration (SOAR)
- Data Exfiltration Technique
- Endpoint Detection & Response
- Detect, Analyze, Respond, Recover

Your Answers:

1	2	3	4	5	6	7	8



TIPS TO IMPROVE

- Understand concepts, not just definitions.
- Practice with real logs and labs.
- Build projects and document everything.
- Stay curious and never stop learning.



SECTION 4: SHORT ANSWER QUESTIONS

- What are the three main components of a SIEM?

- What is the difference between a False Positive and a True Positive?

- Name two common Windows logs monitored by SOC Analysts.

- What are Indicators of Compromise (IOCs)?

- What is the purpose of threat hunting?



BONUS - SCENARIO BASED QUESTION

You received an alert in SIEM showing multiple failed login attempts (Event ID 4625) followed by a successful login (Event ID 4624) from a new IP address.

- What is the possible threat? _____
- What steps will you take next?



KEY TAKEAWAY

The more you practice, the better you get.
Stay consistent, stay focused, and success will follow!



“ A great SOC Analyst doesn't just react to alerts, they understand, investigate, and prevent attacks. ”

